

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Smart Carrer Navigator

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- **Problem Statement :**

- Traditional degree-based hiring is becoming outdated
- Students often lack industry-relevant skills
- Existing platforms only recommend jobs
- No clear guidance on :
 - Why a candidate is rejected
 - Which skills are missing
 - How to improve employability

- **Motivation :**

- Rapid technology evolution creates skill gaps
- Industry demands real-time upskilling
- Need for an AI-driven personalized career guidance system

Objective

Develop an AI-powered framework to :

- Analyze resumes
- Extract skills from Job Descriptions
- Identify missing skills
- Recommend personalized learning paths

Core Features

Resume Parsing using NLP

- Semantic Skill Matching using BERT
- Career Readiness Score
- Personalized Recommendation Engine
- Real-Time Market Analysis

System Modules

- Data Ingestion Layer
- NLP & Named Entity Recognition
- Semantic Embedding Engine
- Similarity Computation
- Recommendation Engine

Technologies Used

- React
- Node.js
- SpaCy
- BERT
- MongoDB

NLP Processing

Skills extracted into:

- Programming Languages
- Frameworks
- Tools/Platforms
- Soft Skills

Semantic Matching

Resume and Job Description converted into vectors.

Similarity Formula : $S(R, J) = (R \cdot J) / (\|R\| \|J\|)$

Where: R = Resume Vector , J = Job Description Vector

Dataset

5,000 resumes

2,000 job descriptions

Domains:

- Web Development
- Data Science
- DevOps
- Cybersecurity

Conclusion

- Smart Career Navigator bridges education and industry requirements
- Uses AI + NLP for intelligent skill gap analysis
- Provides measurable career readiness insights
- Helps students become industry-ready

Future Scope

- GitHub profile analysis
- AI-based mock interview evaluation
- LinkedIn integration
- Multi-language resume parsing
- Voice confidence analysis

Final Line

“Transforming career guidance from static recommendations to intelligent career optimization.”